

No. of Functional Features Included in the Solution

Date	28 June 2026
Team ID	
Project Name	EduGenie
Maximum Marks	5 Marks

Functional Features Overview:

This document outlines the functional features of the EduGenie platform. Each feature is described based on its purpose, implementation, and contribution to the overall AI-powered educational solution, ensuring an effective, personalized, and engaging learning experience for students.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration & Auth	Secure signup/login with hashed passwords and JWT sessions.	app/services/auth_service.py	Done	15%
2	AI Q&A Module	Interactive query answering supporting streaming responses.	app/routes/api.py	Done	20%
3	RAG Document Upload	Contextual educational chat from PDF/DOCX file uploads.	app/services/rag_service.py	Done	20%
4	Concept Explainer	Generates definitions, analogies, and examples for 3 levels.	app/routes/api.py	Done	15%
5	Quiz Generator	Dynamic multiple-choice quiz creator with scoring & PDF export.	app/routes/api.py	Done	15%
6	Progress Analytics	Streaks, badges, and learning path custom roadmaps.	app/services/analytics_service.py	Done	15%

Feature Summary:

Metric	Count / Value
Total Features Planned	6

Total Features Implemented	6
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2 (RAG document upload & analytics dashboard)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Interactive glassmorphism student portal dashboard	Jinja2 dashboard, Q&A template, Quiz panel
2	Backend / Logic	FastAPI server with route handlers	api.py pages.py endpoints
3	Database / Storage	Flat JSON collection data store	users.json, user_queries.json, quizzes.json
4	API / Integration	Google Gemini API & PyPDF2 / docx processing	Gemini 1.5 Pro, local LaMini integration
5	Security / Authentication	HTTPOnly cookies, bcrypt hashing, JWT validation	JWT Auth cookie, bcrypt hashed users.json